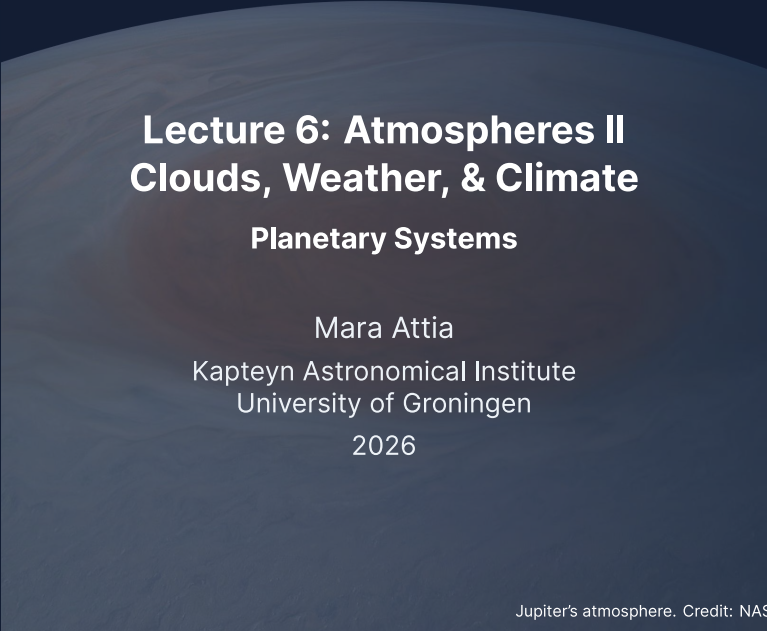


The background of the slide is a high-resolution image of Jupiter's atmosphere, showing swirling cloud patterns in shades of brown, tan, and white. The image is framed by a dark blue border at the top and sides.

Lecture 6: Atmospheres II

Clouds, Weather, & Climate

Planetary Systems

Mara Attia
Kapteyn Astronomical Institute
University of Groningen
2026

Learning objectives

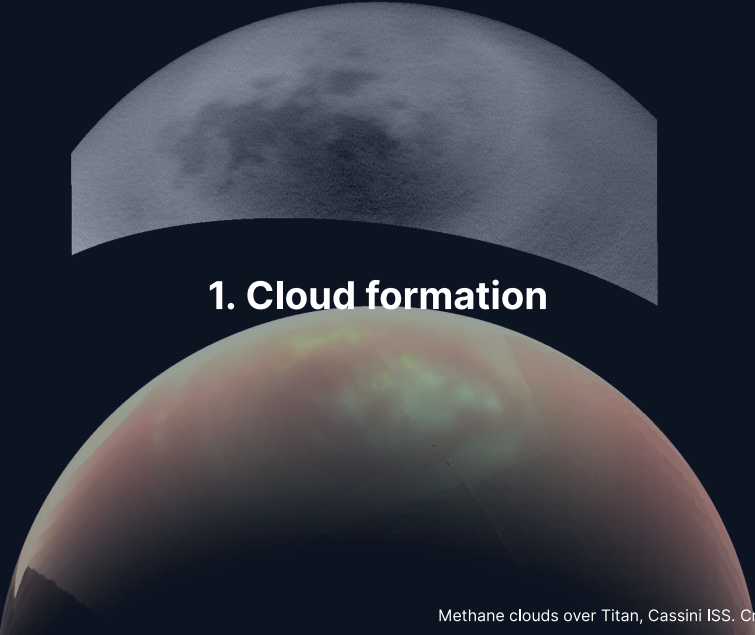
By the end of this lecture, you will be able to:

- Derive the Clausius-Clapeyron equation and predict cloud formation
- Explain circulation using the Coriolis effect and geostrophic balance
- Describe weather phenomena throughout the solar system
- Discuss climate evolution and the faint young Sun paradox
- Explain the carbonate-silicate cycle as a climate thermostat

Recap: Lecture 5

- Hydrostatic equilibrium $dP = -\rho g dz$ and scale height $H = k_B T / (\mu m_u g)$ set atmospheric structure
- Radiative equilibrium and greenhouse warming set surface temperatures ($T_s > T_{eq}$)
- Convection drives the adiabatic lapse rate $\Gamma_d = g/c_p$ in the lower atmosphere

Today: What happens when vapour condenses? How do winds blow? Why has Earth's climate been stable for 4 Gyr?



1. Cloud formation

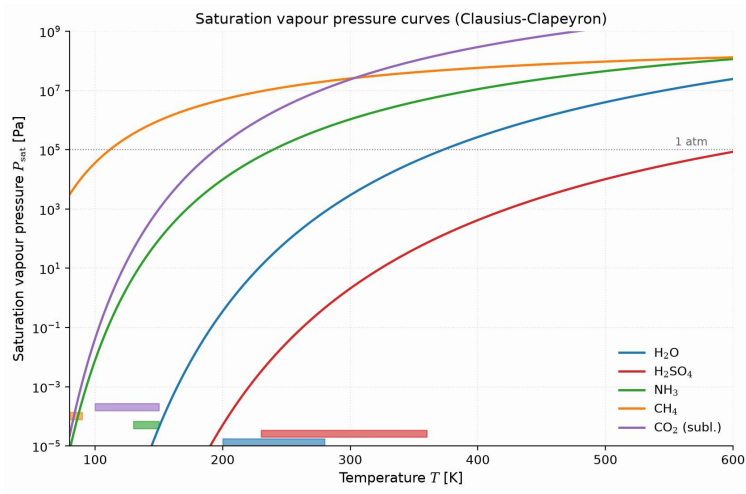
Methane clouds over Titan, Cassini ISS. Credit: NASA/JPL-Caltech/SSI

Saturation and condensation

- Every vapour has a **saturation vapour pressure** $P_{\text{sat}}(T)$
- When actual vapour pressure exceeds P_{sat} : air is **supersaturated**
- $P_{\text{sat}}(T)$ increases roughly exponentially with temperature
- **Relative humidity:**

$$\text{RH} = \frac{P_{\text{vapour}}}{P_{\text{sat}}(T)} \times 100\%$$

- RH = 100%: saturated; RH > 100%: condensation can occur
- Rising air cools $\rightarrow P_{\text{sat}}$ drops $\rightarrow$ reaches saturation $\rightarrow$ clouds



Saturation vapour pressure for the five main condensables. A single equation, evaluated per species, predicts every cloud deck in the solar system. Course figure.

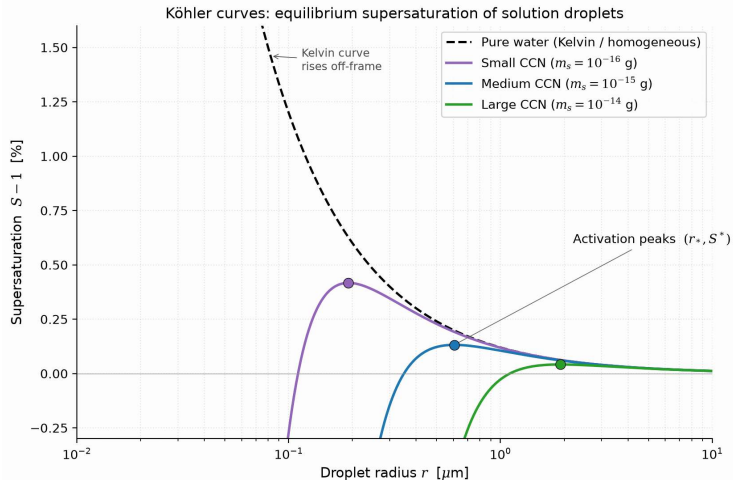
Nucleation: how droplets form

Even in supersaturated air, condensation needs a **seed**.

Homogeneous nucleation: Droplets from vapour alone; requires extreme supersaturation ($RH \gg 100\%$), very rare

Heterogeneous nucleation: Condensation onto pre-existing nuclei (dust, sea salt, soot); activates at only ~0.04–0.4% supersaturation

- Earth: oceanic/biogenic nuclei; Mars: dust; Giant planets: photochemical hazes



Köhler curves: equilibrium supersaturation of solution droplets. Real clouds nucleate heterogeneously, with aerosols lowering the barrier by two orders of magnitude. Course figure.

The lifting condensation level (LCL)



Credit: Eulenjäger, CC BY-SA 3.0

Rising air cools dry adiabatically ($\Gamma_d = g/c_p$) until $P_{\text{vapour}} = P_{\text{sat}}(T)$, i.e. until it reaches its **dew point** T_d .

The **lifting condensation level** (LCL) marks the **cloud base**.

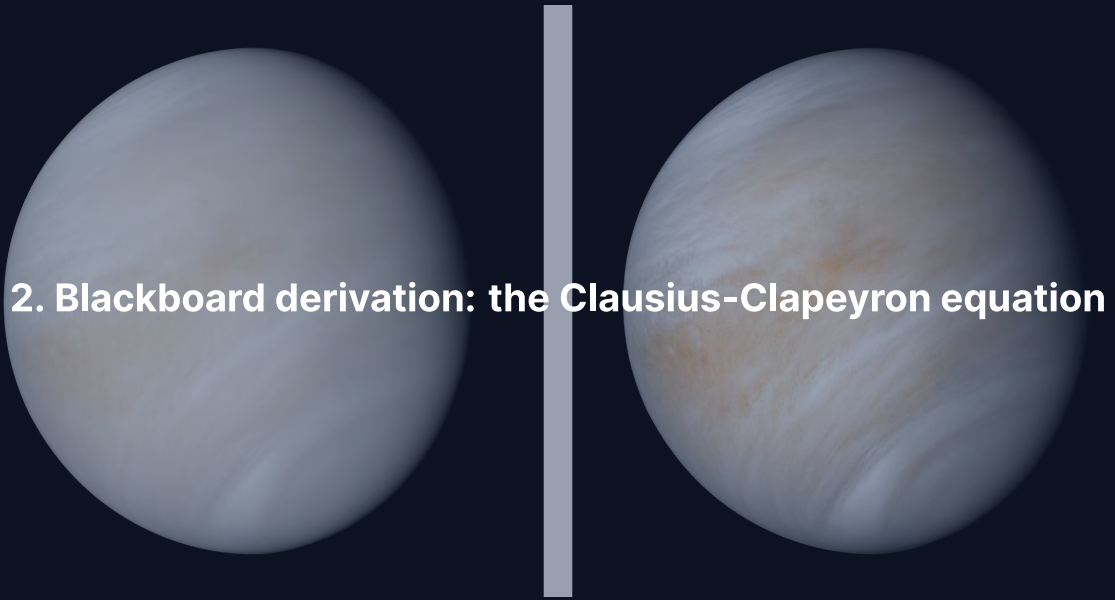
- Above the LCL, latent heat release reduces cooling to the moist adiabatic lapse rate ($\sim 5 \text{ K km}^{-1}$ vs. $\sim 9.8 \text{ K km}^{-1}$ dry)
- Latent heat fuels buoyancy, powering deep convective storms up to the tropopause

Cloud types of the solar system

What condenses depends on temperature and composition:

Body	Cloud species	Cloud base altitude
Earth	H ₂ O	~1–2 km
Venus	H ₂ SO ₄	~48 km
Mars	CO ₂ ice, H ₂ O ice	10–100 km
Titan	CH ₄ , C ₂ H ₆	~8–30 km
Jupiter	NH ₃ , NH ₄ SH, H ₂ O	layered, 0.5–7 bar

The **Clausius-Clapeyron equation** governs every cloud deck.



2. Blackboard derivation: the Clausius-Clapeyron equation

Deriving the Clausius-Clapeyron equation

Goal: Derive the exponential dependence of saturation vapour pressure on temperature from thermodynamic phase equilibrium.

Setup: phase equilibrium.

Along the coexistence curve, liquid and vapour are in equilibrium:

$$g_\ell(T, P) = g_v(T, P)$$

where g is the specific Gibbs free energy.

Moving along the coexistence curve by (dT, dP) :

$$dg_\ell = dg_v$$

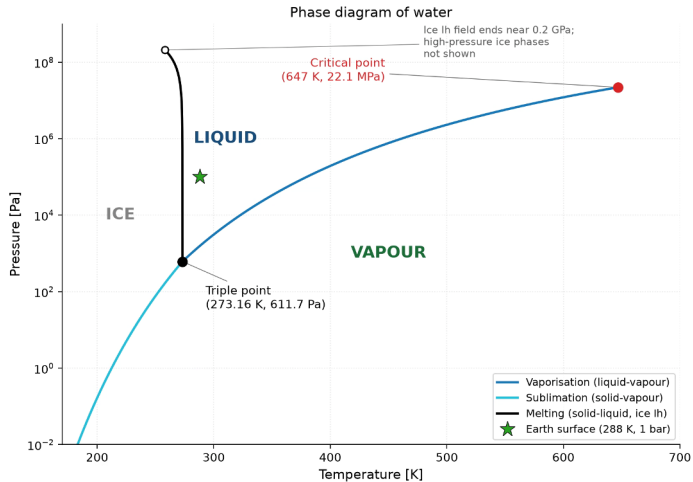
To the board.

The Clausius-Clapeyron equation

$$P_{\text{sat}}(T) = P_{\text{ref}} \exp \left[-\frac{L_v}{R_v} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right]$$

Credit: Derivation follows Catling (2017), *Atmospheric Evolution on Inhabited and Lifeless Worlds*, Ch. 1

- P_{sat} depends **exponentially** on temperature
- The sensitivity is controlled by L_v/R_v (units of K)
- Large $L_v/R_v \rightarrow$ very sensitive to temperature changes
- This equation applies to **any** vapour $\leftrightarrow$ condensed phase transition



Phase diagram of water in P - T space. The liquid-vapour coexistence curve integrates the Clausius-Clapeyron relation from the triple point to the critical point; the solid-vapour curve governs sublimation. Course figure.

Worked example: water on Earth

For water vapour:

L_v	$2.50 \times 10^6 \text{ J kg}^{-1}$
M	$0.018 \text{ kg mol}^{-1}$
$R_v = R^* / M$	$462 \text{ J kg}^{-1} \text{ K}^{-1}$
T_{ref} (triple point)	273 K
P_{ref} (triple point)	611 Pa
L_v / R_v	$\approx 5400 \text{ K}$

At $T = 293 \text{ K}$ (20°C):

$$P_{\text{sat}} = 611 \exp\left[-5400\left(\frac{1}{293} - \frac{1}{273}\right)\right] = 611 \times 3.86 \approx 2360 \text{ Pa}$$

Measured value: 2300 Pa; excellent agreement.

Condensing species in the solar system

Species	L_v (kJ kg ⁻¹)	R_v (J kg ⁻¹ K ⁻¹)	L_v/R_v (K)	T_{cond} (K)	Where
H ₂ O	2500	462	5400	200–280	Earth, Mars
H ₂ SO ₄	540	85	6400	230–360	Venus
NH ₃	1370	488	2800	130–150	Jupiter, Saturn
CH ₄	510	519	980	80–90	Titan, Uranus, Neptune
CO ₂	571	189	3020	100–150	Mars

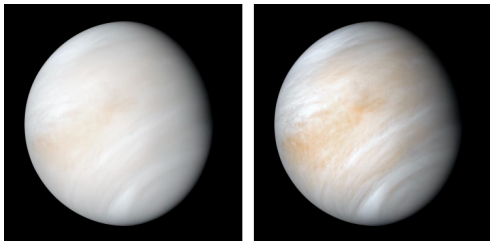
Steep L_v/R_v curves sharpen cloud bases (Venus); flat curves broaden decks (Titan).



3. Clouds of the solar system

Water-ice clouds over Mars, Mars Odyssey THEMIS (2025). Credit: NASA/JPL-Caltech/ASU

Venus: sulfuric acid clouds

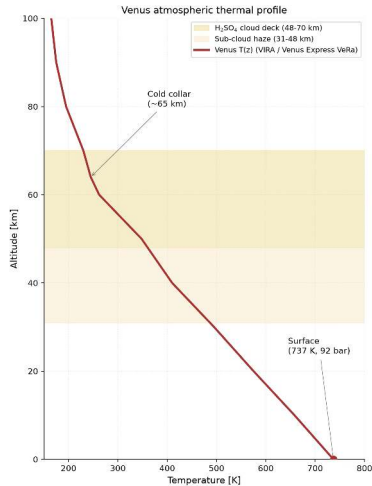


Credit: NASA/JPL-Caltech (1974)

- Concentrated H_2SO_4 clouds span 48–70 km altitude
- Photochemistry from outgassed SO_2 and photolytic O:

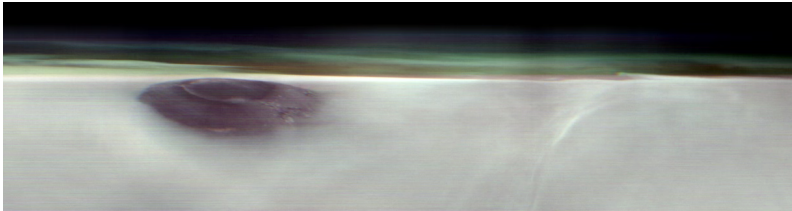


- Unidentified UV absorber creates upper-cloud tracking features
- Sub-cloud haze extends down to ~31 km



Venus thermal structure, VIRA / Venus Express VeRa. Concentrated H_2SO_4 clouds form a 22 km deck at 48–70 km altitude where Clausius-Clapeyron allows them. Course figure.

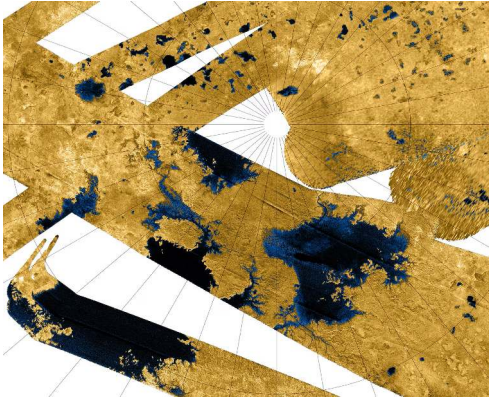
Mars: dust and ice clouds



Water-ice clouds over Tharsis limb. Credit: NASA/JPL-Caltech/ASU (Mars Odyssey THEMIS, 2 May 2025)

- CO_2 **ice**: mesospheric, 50–100 km, where T falls below the local CO_2 frost point (148 K at the 6 mbar surface, lower aloft)
- H_2O **ice**: 10–30 km over Tharsis volcanoes
- Mineral dust acts as condensation nuclei
- Radiative heating drives positive dust feedback

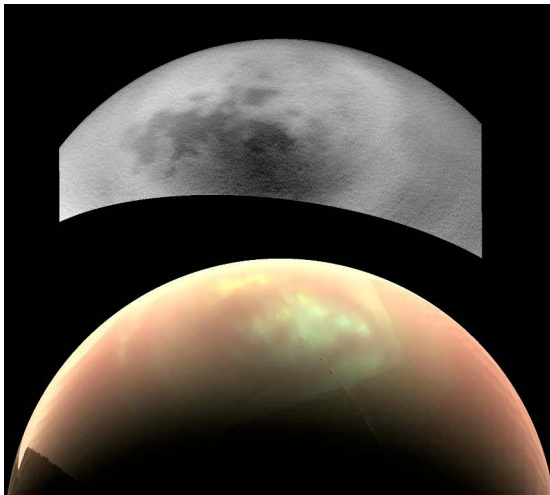
Titan's methane cycle: an Earthlike analogue



Credit: NASA/JPL-Caltech/ASI/USGS (Cassini)

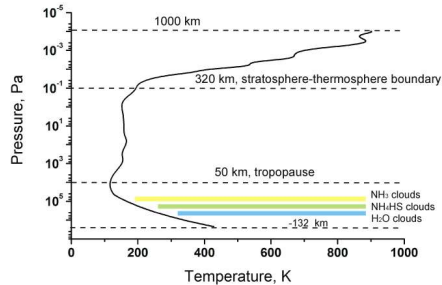
Active hydrological cycle with CH_4 near its triple point:

- Surface at 94 K and 1.5 bar supports liquid methane seas
- **Kraken Mare:** ~400,000 km², locally >100 m deep
- Methane rain carves river channels and fills polar lakes
- Driven by slow 30-year seasonal solar cycling



Methane clouds at Titan's mid-southern latitudes, Cassini ISS. Methane rain at 94 K drives the only active hydrological cycle beyond Earth with polar convective storms. Credit: NASA/JPL-Caltech/SSI.

Giant planets: layered cloud structure



Credit: Wikimedia, CC BY-SA 3.0

Vertically layered, predicted by Clausius-Clapeyron:

1. NH₃ **ice** (top): 130–150 K, 0.5–1 bar
2. NH₄SH: 200–240 K, 2–3 bar (NH₃ + H₂S)
3. H₂O **ice/liquid** (deep): 270–300 K, 5–7 bar

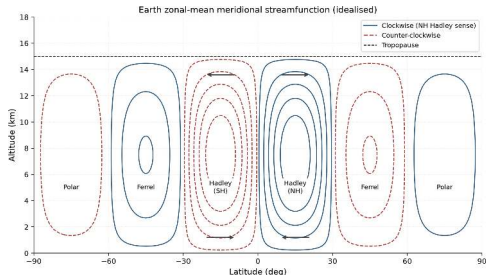
Latent heat release from deep water clouds fuels moist convection.

A satellite image of Earth from space, showing a wide view of the planet's surface. The image captures the curvature of the Earth, with a clear horizon line. The atmosphere is visible as a thin blue layer. Large, white, swirling cloud patterns are prominent, particularly in the upper right and lower left quadrants, indicating high-level cloud formations like jet-stream cirrus. The landmasses are visible in shades of brown and green, with some white areas representing ice or snow. The overall scene is a high-contrast, high-resolution view of the Earth's atmosphere and surface features.

4. Atmospheric dynamics

Jet-stream cirrus from orbit. Credit: NASA/JSC

The Hadley cell



Credit: Course figure

1. Solar heating at the equator drives buoyant rising air
2. Air flows poleward aloft, cooling radiatively
3. Sinking air at $\sim 30^\circ$ latitude creates subtropical desert belts
4. Equatorward surface return flow forms the trade winds

The primary tropical heat transport loop from equator to mid-latitudes.

The Coriolis effect

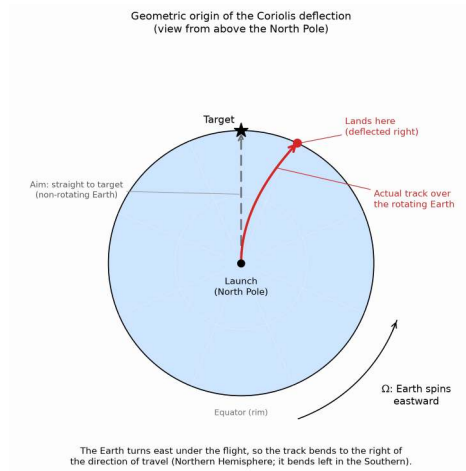
On a rotating planet, moving air is deflected right in the Northern Hemisphere, left in the Southern.

The **Coriolis parameter**:

$$f = 2\Omega \sin \phi$$

- Earth: $\Omega = 7.29 \times 10^{-5} \text{ rad s}^{-1}$
- Mid-latitudes ($\phi = 45^\circ$): $f \approx 1.03 \times 10^{-4} \text{ s}^{-1}$
- Equator ($\phi = 0^\circ$): $f = 0$, so no Coriolis deflection

Explains why winds circulate around pressure centres instead of blowing directly from high to low.



Coriolis deflection of a polar launch, viewed from above the North Pole. The deflection is an apparent force caused by the planet rotating beneath a straight path. Course figure.

The Rossby number

Does planetary rotation dominate a given atmospheric flow?

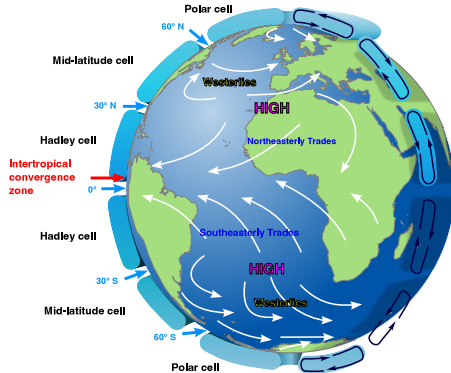
$$Ro = \frac{U}{fL}$$

$Ro \ll 1$: Rotation dominates: large-scale circulation, jet streams (Earth:
 $U \sim 10 \text{ m s}^{-1}$, $L \sim 1000 \text{ km} \rightarrow Ro \sim 0.1$)

$Ro \gg 1$: Rotation unimportant: small-scale convective plumes, tornadoes

The Rossby number sets the dynamic regime and the number of circulation cells.

Circulation cells and rotation rate

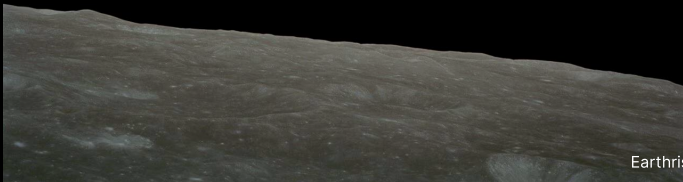


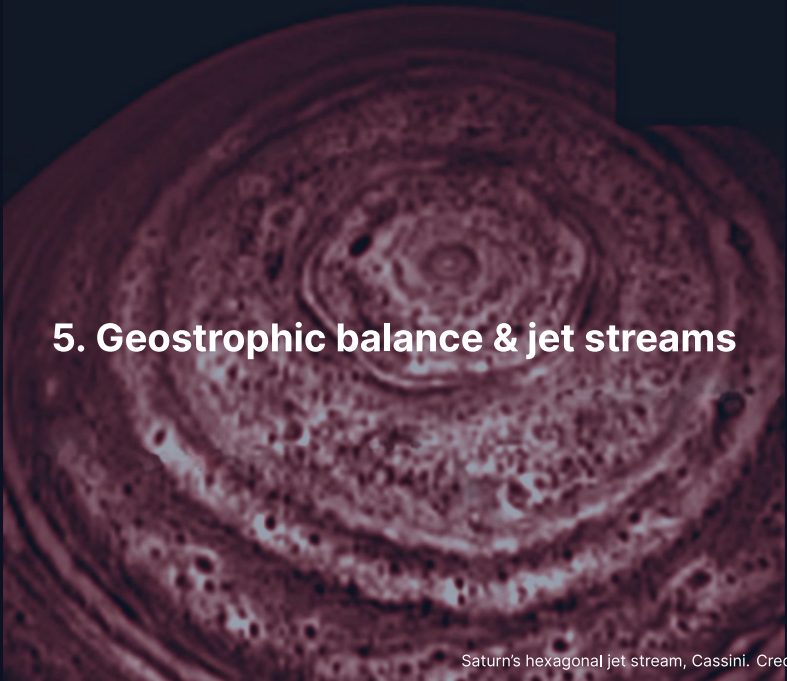
Credit: Wikimedia Commons, public domain

Planetary rotation rate governs the number of circulation cells:

- **Slow rotators** (Venus, Titan): single equator-to-pole Hadley cell per hemisphere ($Ro \gg 1$)
- **Moderate rotators** (Earth): three cells (Hadley, Ferrel, polar)
- **Fast rotators** (Jupiter, Saturn): multiple cells produce banded jets

Break



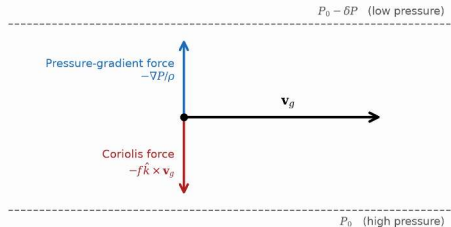
A false-color image of Saturn's atmosphere, showing a prominent hexagonal jet stream. The hexagon is a large, six-sided cloud pattern with a central vortex. The surrounding atmosphere shows various other cloud features and bands. The image is in shades of purple, blue, and white.

5. Geostrophic balance & jet streams

Saturn's hexagonal jet stream, Cassini. Credit: NASA/JPL-Caltech/SSI

Geostrophic balance

Geostrophic balance (Northern Hemisphere)



At large scales ($Ro \ll 1$), Coriolis force balances pressure gradient:

$$f \hat{k} \times \mathbf{v}_g = -\frac{1}{\rho} \nabla P$$

$$u_g = -\frac{1}{f\rho} \frac{\partial P}{\partial y}, \quad v_g = \frac{1}{f\rho} \frac{\partial P}{\partial x}$$

Geostrophic winds blow **parallel to isobars**, rotating cyclonically around lows and anticyclonically around highs.

Jet streams and the thermal wind

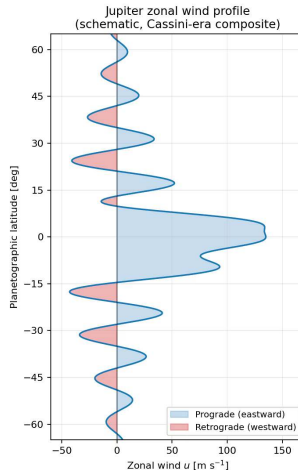


Credit: NASA JSC, public domain

Horizontal temperature gradients drive vertical wind shear through the thermal wind relation:

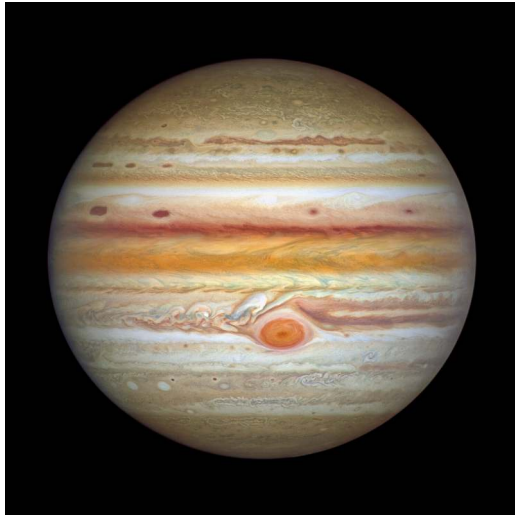
$$\frac{\partial \mathbf{v}_g}{\partial z} \propto \hat{\mathbf{k}} \times \nabla T$$

- Steep equator-to-pole ∇T drives fast winds aloft ($\sim 30\text{--}70 \text{ m s}^{-1}$ on Earth)
- Earth hosts subtropical ($\sim 30^\circ$) and polar ($\sim 60^\circ$) jet streams
- Upper-tropospheric jets steer mid-latitude weather systems across the globe



Jupiter's cloud-top zonal winds, schematic after García Melendo & Sánchez-Lavega (2001).
Alternating prograde and retrograde jets reach peak speeds of $\sim 180 \text{ m s}^{-1}$ and extend thousands of kilometres deep. Course figure.

Giant planet banding



Credit: NASA/ESA/STScI (HST OPAL)

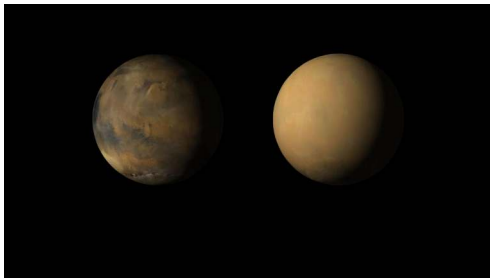
Planetary rotation organises giant-planet atmospheres into alternating bright zones and dark belts:

- **Zones (light):** anticyclonic upwelling, cool ammonia ice cloud tops
- **Belts (dark):** cyclonic downwelling exposing warmer lower layers
- Zonal jets flank band boundaries (Jupiter $\sim 180 \text{ m s}^{-1}$, Saturn $\sim 400 \text{ m s}^{-1}$); decay depth $\sim 1800 \text{ km}$

A close-up, curved view of Jupiter's atmosphere, showing the Great Red Spot as a large, reddish-orange oval storm system. The surrounding atmosphere is a mix of light and dark brownish-grey bands, indicating different cloud layers and chemical compositions. The curvature of the planet is visible at the top and bottom edges of the frame.

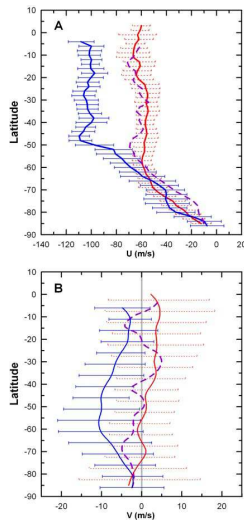
6. Weather & storms in the solar system

Mars: dust storms



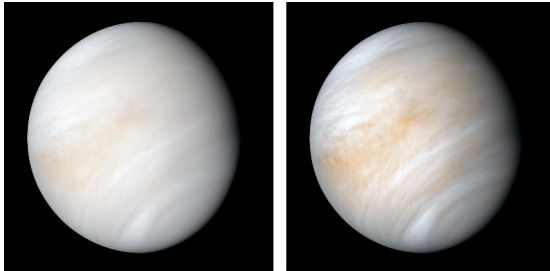
Credit: NASA/JPL-Caltech/MSSS

- Dust lofted to 10–30 km can grow into global dust storms
- **Positive feedback loop:** dust absorbs sunlight → heats air → strengthens winds → lofts more dust
- Seasonal CO₂ condensation freezes 25–30% of the atmosphere onto polar caps, swinging the surface pressure between ~5 and 7 mbar



Venus cloud-level winds from VIRTIS on Venus Express: zonal (A) and meridional (B) speed against latitude. Cloud-top zonal winds reach $\sim 105 \text{ m s}^{-1}$, demonstrating that the entire cloud deck super-rotates 60 times faster than the surface. Sánchez-Lavega et al. (2008).

Venus super-rotation: thermal tides



Credit: NASA/JPL-Caltech (Mariner 10)

Cloud-top winds circle Venus in ~4 days ($\sim 100 \text{ m s}^{-1}$), 60 times faster than the surface.

- **Puzzle:** sustaining super-rotation requires transporting momentum upward and equatorward
- **Thermal tides:** solar heating generates waves that pump momentum upward
- **Planetary waves:** large-scale waves carry momentum toward the equator

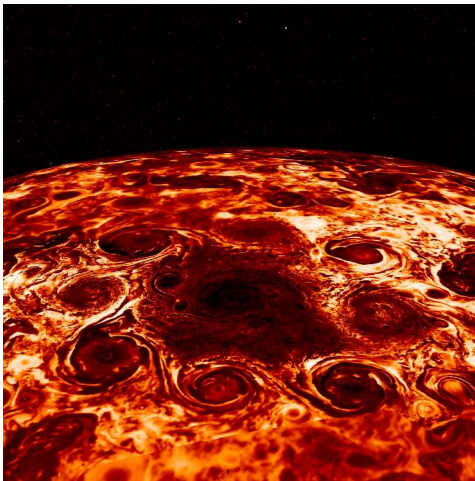
Source: Sánchez-Lavega et al. (2017)

Jupiter: the Great Red Spot



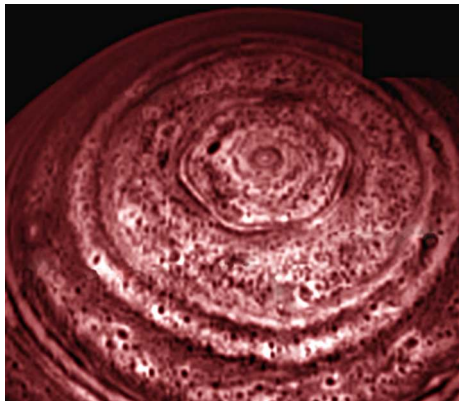
Credit: NASA/JPL-Caltech/SwRI/MSSS (Juno)

- Anticyclonic vortex larger than Earth; continuous records since ~1830
- Peripheral winds reach $\sim 120 \text{ m s}^{-1}$ between opposing zonal jets
- Sustained by merging with smaller vortices and deep H_2O latent heat release
- Slowly shrinking in longitudinal extent over the past century



Polar cyclone cluster at Jupiter's north pole from Juno JIRAM. A central cyclone is encircled by eight circumpolar cyclones in a stable octagonal pattern, demonstrating coherent polygonal vortex dynamics in giant planet atmospheres. Credit: NASA/JPL-Caltech/SwRI/ASI/INAF/JIRAM.

Saturn: the hexagonal jet stream



Credit: NASA/JPL-Caltech/SSI (Cassini, 2013)

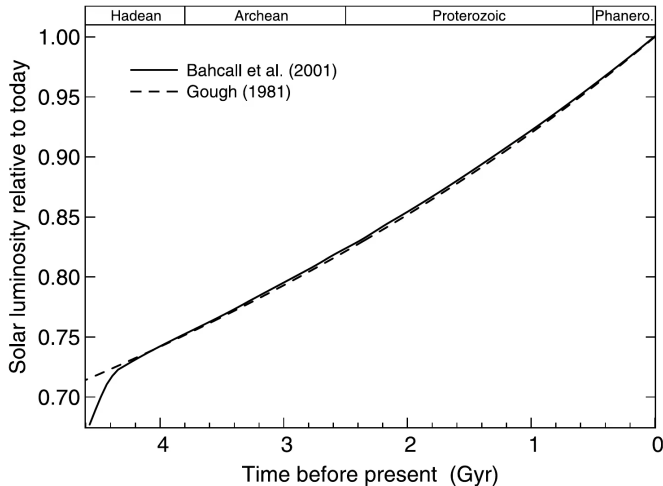
- Persistent hexagonal pattern at $\sim 78^\circ\text{N}$, diameter $\sim 30,000$ km
- Imaged by Voyager (1980–81) and Cassini
- Stationary Rossby wave with azimuthal wavenumber $m = 6$ resonant with the zonal jet
- Saturn also erupts episodic Great White Storms every ~ 30 years (last in 2010)



Neptune's Great Dark Spot, Voyager 2 (1989). Neptune radiates $\sim 2.6\times$ more energy than it receives from the Sun, with internal heat driving $\sim 400 \text{ m s}^{-1}$ winds despite negligible solar heating. Credit: NASA/JPL-Caltech.

An artist's impression of a Snowball Earth, showing a frozen planet with a thin blue ring of liquid water at the equator and a dark blue polar region. The planet is set against a dark blue background with faint stars.

7. Climate evolution & the faint young Sun

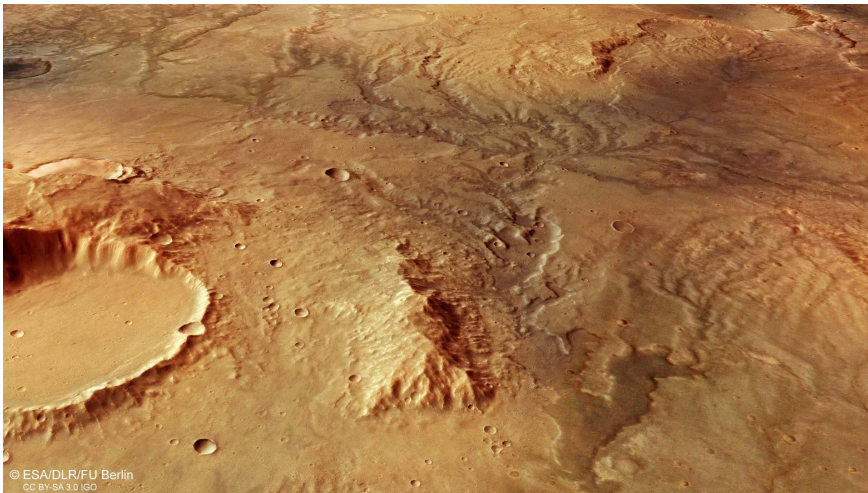


Solar luminosity across geologic time follows $L(t)/L_{\odot} \approx [1 + \frac{2}{5}(1 - t/t_{\odot})]^{-1}$; at $t = 0$, the Sun was ~30% fainter ($0.71 L_{\odot}$) than today. Reproduced from Feulner (2012).

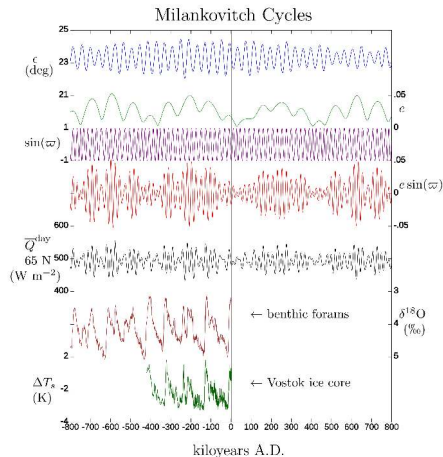
The faint young Sun paradox and solutions

A 30% fainter early Sun implies $T_{\text{eff}} \approx 234$ K, yet geological records require liquid water for >4 Gyr.

- **Geological evidence:** zircons (4.4 Ga), pillow basalts (3.8 Ga), and stromatolites (3.5 Ga) prove ancient oceans
- **Paradox:** with today's atmosphere, early Earth would have been globally frozen
- **Solutions:** stronger greenhouse from elevated CO_2 and biogenic CH_4 , plus lower continental albedo



Noachian valley networks, Mars Express HRSC. Fluvial networks require warm, wet conditions on early Mars (>3.7 Ga), an open paradox under the faint young Sun since pure CO_2 condenses into high-albedo clouds. Credit: ESA/DLR/FU Berlin, CC BY-SA 3.0 IGO.

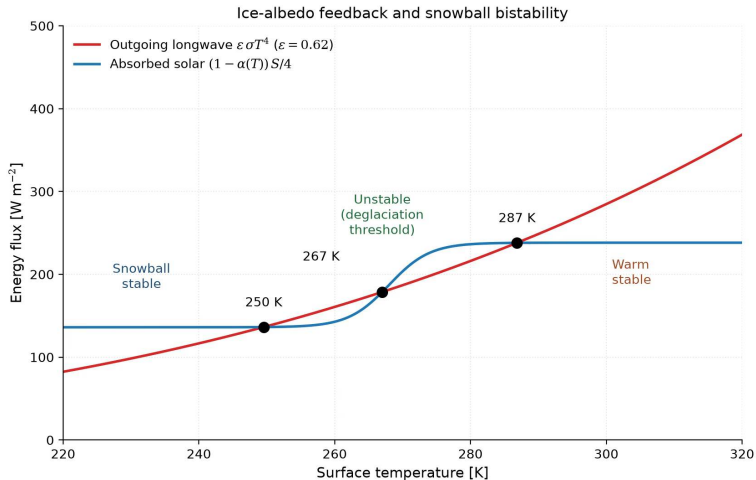


Milankovitch cycles over 800 kyr: obliquity, eccentricity, and precession modulate high-latitude summer insolation, pacing glacial cycles observed in ice-core and sediment records, though the 100 kyr rhythm requires non-linear climate feedbacks. Credit: Wikimedia Commons, CC BY-SA.

Climate feedbacks and the ice-albedo instability

Atmospheric feedbacks determine whether a planetary climate remains stable or runs away.

- **Water vapour feedback:** warming increases vapour, doubling CO_2 warming (runaway on Venus)
- **Cloud feedback:** competing solar reflection (cooling) and infrared trapping (warming)
- **Ice-albedo feedback:** cooling expands reflective ice; past $\sim 30^\circ$ latitude, runaway cooling triggers a snowball state



Energy-balance bistability of the ice-albedo feedback. Stable crossings at 250 K and 287 K bracket an unstable threshold at 267 K, separating warm and snowball branches. Course figure.

Snowball Earth

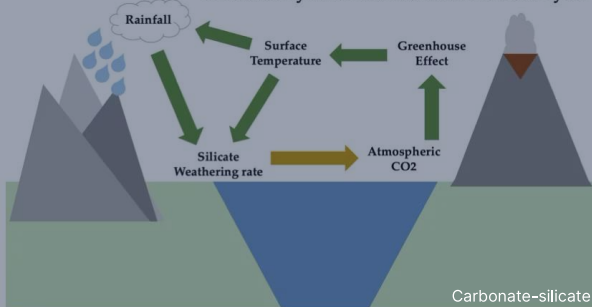
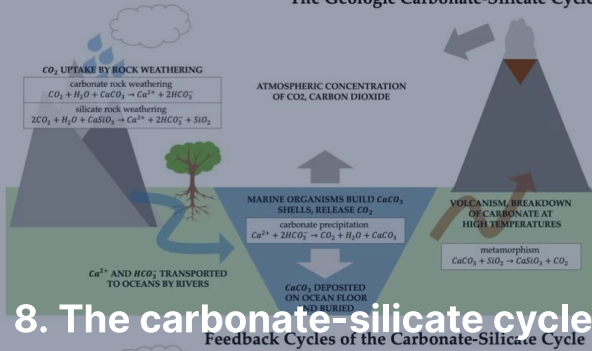


Credit: Oleg Kuznetsov, CC BY-SA 4.0

- Two global glaciations in the Neoproterozoic (~717 and 635 Ma) with tropical glacial deposits
- Surface temperatures dropped below -50°C
- Global ice cover shut down silicate weathering sinks
- Volcanic CO_2 accumulated unchecked over ~10–100 Myr until greenhouse warming melted the ice

Source: Hoffman et al. (2017)

The Geologic Carbonate-Silicate Cycle

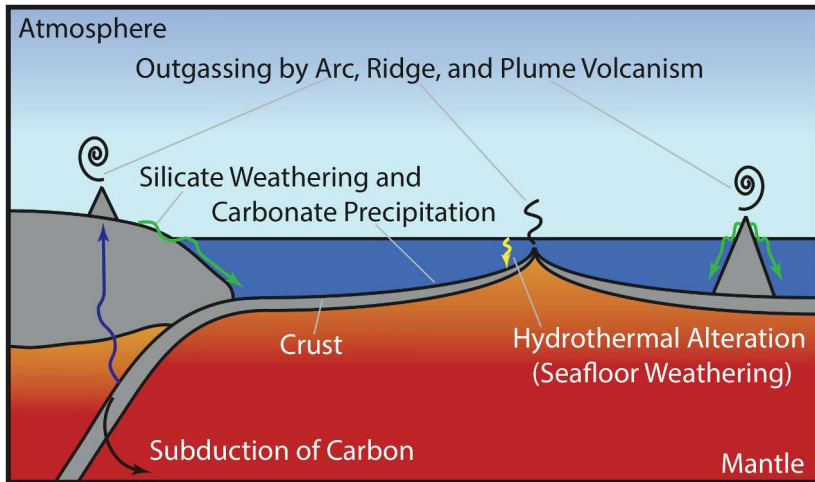


The Urey reaction

Silicate weathering draws atmospheric CO_2 dissolved in rainwater into mineral carbonates:

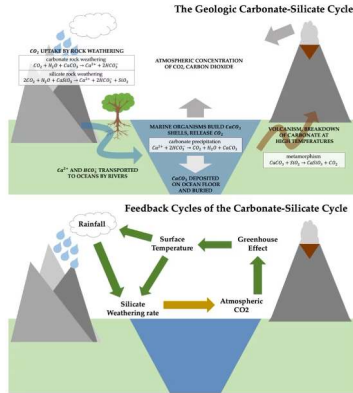


- Atmospheric CO_2 dissolves in rain forming carbonic acid (H_2CO_3)
- Acid weathers surface silicates, washing Ca^{2+} and HCO_3^- to the oceans
- Marine organisms and abiotic processes precipitate carbonate sediments (CaCO_3)
- Net effect: moves carbon from the atmosphere-ocean system into the crust, until subduction and volcanism return it



Carbonate-silicate cycle on the modern Earth. Atmospheric CO_2 dissolves in rainwater, weathers surface silicates into oceanic carbonates, and plate-tectonic subduction returns metamorphic CO_2 to the atmosphere via arc and plume volcanism. Credit: Foley (2024).

Volcanic outgassing: the carbon source



Credit: Wikimedia, CC BY-SA 4.0

- Plate tectonics subducts carbonate-rich seafloor into the mantle
- High T and P decompose carbonates, releasing metamorphic CO_2
- **Volcanic outgassing** returns CO_2 to the atmosphere
- Long-term steady state balances weathering sinks against volcanic sources

The negative feedback thermostat

Silicate weathering provides temperature-dependent negative feedback:

- **Warming:** higher rainfall and reaction rates accelerate weathering, drawing down CO_2 and cooling the planet
- **Cooling:** reduced weathering allows volcanic CO_2 to accumulate, strengthening the greenhouse

Operates on ~0.5 Myr timescales, maintaining liquid water for >4 Gyr.

Quantifying the weathering feedback

Silicate weathering rates increase exponentially with temperature to stabilise climate:

$$W(T) = W_0 \exp\left(\frac{T - T_0}{T_e}\right)$$

- With $T_e \approx 10$ K, doubling weathering requires ~ 7 K warming
- Climate perturbations damp on a ~ 0.5 Myr timescale
- Feedback requires liquid water and active volcanism to operate

Source: Walker et al. (1981); Pierrehumbert (2010)

Venus & Mars: failed thermostats

The thermostat requires liquid water and volcanism:

Venus (lost sink):

- Runaway greenhouse boiled oceans
- No water → no weathering sink
- Volcanic CO₂ reached 92 bar

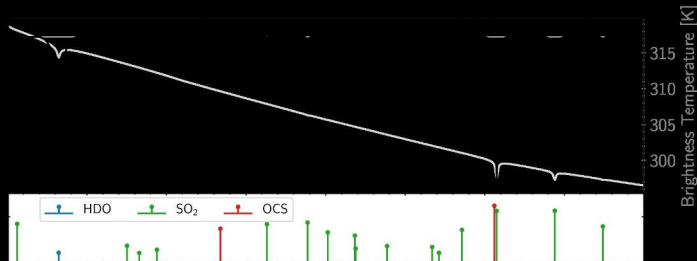
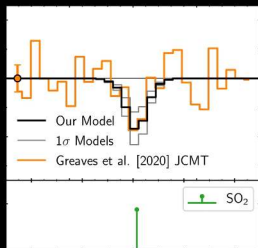
Mars (lost source):

- Volcanism ceased ~3 Ga
- No volcanic return path
- Weathering drawdown and escape left ~6 mbar

Habitability requires active geological cycling, not sunlight alone.

The image features two spheres side-by-side against a dark blue background. The sphere on the left is a realistic representation of Mars, showing various shades of brown, tan, and blue, with visible surface features like craters and polar ice caps. The sphere on the right is a uniform, hazy, light-brown color, representing the planet during a global dust storm. Centered over the space between these two spheres is the text '9. Recent advances & outlook' in a bold, white, sans-serif font.

9. Recent advances & outlook



The reported 267 GHz Venus phosphine feature is fit by mesospheric SO₂, a controversy that NASA's DAVINCI descent probe will test in situ. Lincowski et al. (2021).

Summary

1. Clausius-Clapeyron equation predicts saturation vapour pressure and cloud bases
2. Planetary rotation sets circulation cells, from single Hadley loops to banded jets
3. Geostrophic balance and thermal wind govern large-scale winds and storms
4. Faint young Sun paradox is resolved by greenhouse warming and lower albedo
5. Carbonate-silicate weathering provides a ~ 0.5 Myr climate thermostat

Next: Lecture 7

Questions?

Next: Lecture 7, Planetary surfaces: geology, geomorphology & geophysics

Mon 28 Sep, 15:00–17:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

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